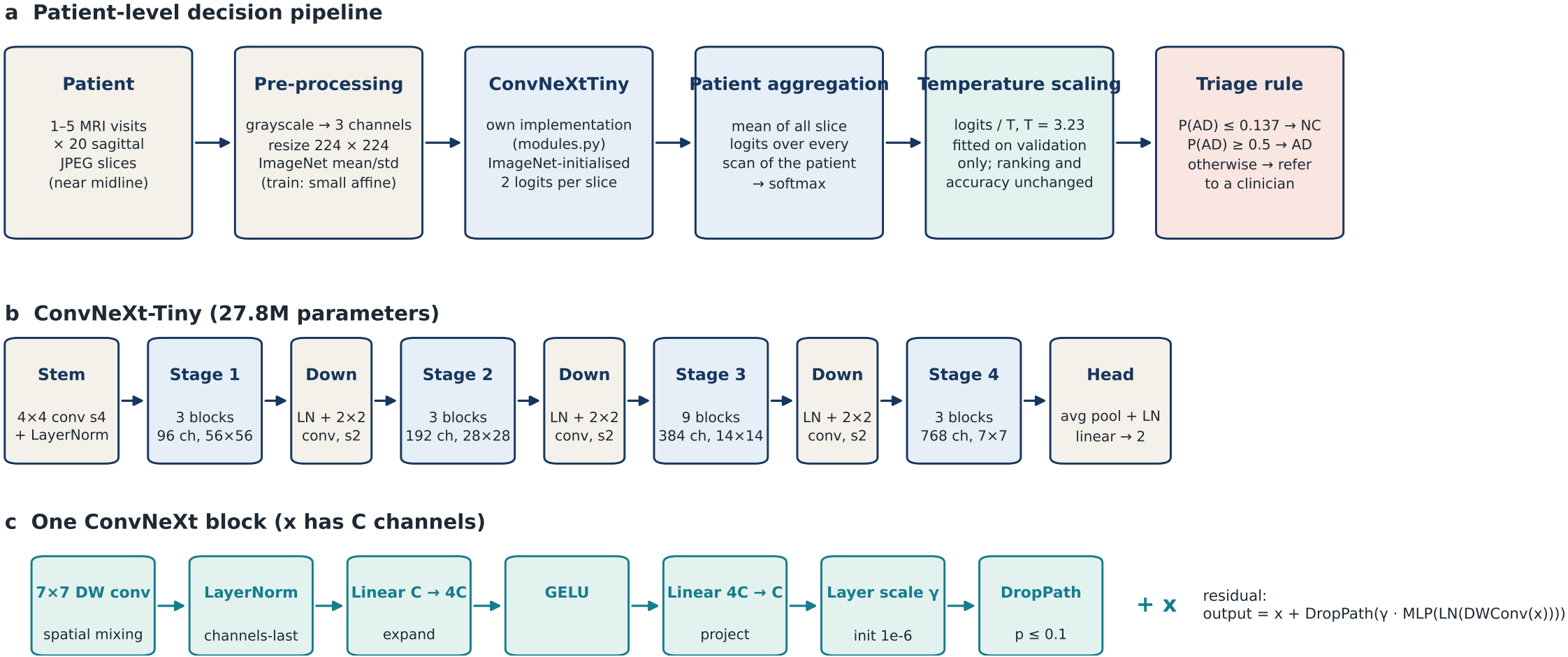
a Patient-level decision pipeline
Patient
1–5 MRI visits × 20 sagittal JPEG slices (near midline)
Pre-processing
grayscale → 3 channels
resize 224 × 224
ImageNet mean/std
(train: small affine)
ConvNeXtTiny
own implementation (modules.py)
ImageNet-initialised
2 logits per slice
Patient aggregation
mean of all slice logits over every scan of the patient → softmax
Temperature scaling
logits / T, T = 3.23
fitted on validation only; ranking and accuracy unchanged
Triage rule
P(AD) ≤ 0.137 → NC
P(AD) ≥ 0.5 → AD
otherwise → refer to a clinician
b ConvNeXt-Tiny (27.8M parameters)
Stem
4×4 conv s4 + LayerNorm
Stage 1
3 blocks
96 ch, 56×56
Down
LN + 2×2 conv, s2
Stage 2
3 blocks
192 ch, 28×28
Down
LN + 2×2 conv, s2
Stage 3
9 blocks
384 ch, 14×14
Down
LN + 2×2 conv, s2
Stage 4
3 blocks
768 ch, 7×7
Head
avg pool + LN
linear → 2
c One ConvNeXt block (x has C channels)
7×7 DW conv
spatial mixing
LayerNorm
channels-last
Linear C → 4C
expand
GELU
Linear 4C → C
project
Layer scale γ
init 1e-6
DropPath
p ≤ 0.1
+ x
residual:
output = x + DropPath(γ · MLP(LN(DWConv(x))))